

Bölüm 21 Zaman Bağı. pot

Fourier Dalga Denk $\Rightarrow$ Schr. Denk $= i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2\mu} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = E\Psi$

$$\Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = E\Psi \Rightarrow \frac{\partial \Psi}{\Psi} = \frac{E dt}{i\hbar} \Rightarrow \ln \Psi = \frac{Et}{i\hbar} + \ln C$$

$$\Rightarrow \frac{\ln \Psi}{\ln C} = \frac{Et}{i\hbar} \Rightarrow \Psi = C e^{-iEt/\hbar}$$

$$\Psi(t) = \sum_n c_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle$$

$$H = H_0 + \lambda V(t), t > 0$$

$$t=0 \text{ da } H_0 \phi_n^0 = E_n^0 \phi_n^0$$

$$i\hbar \frac{\partial \Psi}{\partial t} = (H_0 + \lambda V(t)) \Psi(t)$$

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \left(\sum_n c_n(t) e^{-iE_n^0 t/\hbar} \right) = (H_0 + \lambda V(t)) \left(\sum_n c_n(t) e^{-iE_n^0 t/\hbar} \right)$$

$$\Rightarrow i\hbar \left\{ \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} - \sum_n c_n(t) E_n^0 e^{-iE_n^0 t/\hbar} \right\} = \left(\sum_n c_n(t) H_0 e^{-iE_n^0 t/\hbar} + \lambda \sum_n c_n(t) V(t) e^{-iE_n^0 t/\hbar} \right) |\phi_n\rangle$$

$$\Rightarrow i\hbar \left\{ \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle - \sum_n c_n(t) E_n^0 e^{-iE_n^0 t/\hbar} |\phi_n\rangle \right\} = \sum_n c_n(t) e^{-iE_n^0 t/\hbar} E_n^0 |\phi_n\rangle + \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} V(t) |\phi_n\rangle$$

$$\langle \phi_m | \Rightarrow i\hbar \left\{ \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle - \sum_n c_n(t) e^{-iE_n^0 t/\hbar} E_n^0 \langle \phi_m | \phi_n \rangle \right\} = \sum_n c_n(t) e^{-iE_n^0 t/\hbar} E_n^0 \langle \phi_m | \phi_n \rangle + \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$i\hbar \left\{ \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle \right\} = + \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\dot{c}_m(t) e^{-iE_m^0 t/\hbar} = \frac{\lambda}{i\hbar} \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

veya $\dot{c}_m(t) e^{-iE_m^0 t/\hbar} = \frac{\lambda}{i\hbar} \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$

$$c_n(t) = \lambda^0 c_n^0 + \lambda^1 c_n^1 + \lambda^2 c_n^2 \Rightarrow$$

$$\dot{c}_m(t) e^{-iE_m^0 t/\hbar} = \frac{\lambda}{i\hbar} \sum_n (c_n^0 + \lambda c_n^1 + \lambda^2 c_n^2) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\dot{c}_m^0 + \lambda \dot{c}_m^1 + \lambda^2 \dot{c}_m^2 = \frac{\lambda}{i\hbar} \sum_n (c_n^0 + \lambda c_n^1 + \lambda^2 c_n^2) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^0 \Rightarrow \dot{c}_m^0(t) = 0 \Rightarrow c_n^0 = \text{sb} = \delta_{mn} \quad m=n \Rightarrow$$

$$\lambda^1 \Rightarrow \dot{c}_m^1(t) = \frac{1}{i\hbar} \sum_n c_n^0 e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^2 \Rightarrow \dot{c}_m^2(t) = \frac{1}{i\hbar} \sum_n c_n^1 e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\dot{c}_m^0(t) = 0 \Rightarrow H \text{ e } \rho \text{ si } 0 \text{ olur}$$

$$E_n^0 - E_m^0 = \omega_{nm}$$

$$\dot{c}_m^1(t) = \frac{1}{i\hbar} e^{-i\omega_{nm}t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\frac{d c_m^1(t)}{dt} = \frac{1}{i\hbar} e^{-i\omega_{nm}t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$= c_m^1(t) = \frac{1}{i\hbar} \int dt e^{-i\omega_{nm}t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\text{Schrödinger} = i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = E\Psi$$

$$\Rightarrow i\hbar \frac{\partial \Psi}{\partial t} = E\Psi \Rightarrow \frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar} E\Psi \Rightarrow \ln \Psi = \frac{1}{i\hbar} Et + \ln C \Rightarrow \frac{\partial \Psi}{\partial t} = \frac{1}{i\hbar} E\Psi \Rightarrow \Psi = C e^{-iEt/\hbar} = \sum_n c_n e^{-iE_n t/\hbar} |\phi_n\rangle$$

$$H = H_0 + \lambda V(t) \Rightarrow H = H_0 + \lambda V(t)$$

$$i\hbar \frac{d}{dt} \left(\sum_n c_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle \right) = (H_0 + \lambda V(t)) \left(\sum_n c_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle \right)$$

$$i\hbar \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle + \sum_n c_n(t) e^{-iE_n^0 t/\hbar} E_n^0 |\phi_n\rangle = \sum_n c_n(t) e^{-iE_n^0 t/\hbar} H_0 |\phi_n\rangle + \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} V(t) |\phi_n\rangle$$

$$\langle \phi_m | \Rightarrow i\hbar \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle + \sum_n c_n(t) e^{-iE_n^0 t/\hbar} E_n^0 \langle \phi_m | \phi_n \rangle = \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | H_0 | \phi_n \rangle + \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$i\hbar \sum_n \dot{c}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle = \lambda \sum_n c_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\sum_n \dot{c}_n(t) \langle \phi_m | \phi_n \rangle = \frac{\lambda}{i\hbar} \sum_n c_n(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\dot{c}_m(t) = \frac{\lambda}{i\hbar} \sum_n c_n(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^0 c_m^0(t) + \lambda c_m^1(t) + \lambda^2 c_m^2(t) = \frac{\lambda}{i\hbar} \sum_n \left(\lambda^0 c_n^0(t) + \lambda^1 c_n^1(t) + \lambda^2 c_n^2(t) \right) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^0 \Rightarrow c_m^0(t) = 0 \Rightarrow c_n^0(t) = \delta_{nt} = \delta_{nm}$$

$$\lambda^1 \Rightarrow \dot{c}_m^1(t) = \frac{1}{i\hbar} \sum_n c_n^0(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^2 \Rightarrow \dot{c}_m^2(t) = \frac{1}{i\hbar} \sum_n c_n^1(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\lambda^1 \Rightarrow \frac{d c_m^1(t)}{dt} = \frac{1}{i\hbar} \sum_n \underbrace{\delta_{nm}}_1 e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$\frac{d c_m^1(t)}{dt} = \frac{1}{i\hbar} e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle$$

$$c_m^1(t) = \int \frac{1}{i\hbar} e^{-i\omega_{nm}t/\hbar} \langle \phi_m | V(t) | \phi_n \rangle dt$$

